

OB1 Protocol Specification: Open Brain Static Retrieval over MCP

1. System Identity & Overview

The **OB1 (Open Brain / Obi-Wan)** protocol defines the formal Model Context Protocol (MCP) server contract for deterministically querying and updating persistent memory across the federated #d.u.m.b.a.s.s. network.

Unlike dynamic conversational search, OB1 treats memory as an immutable, time-sliced ledger with cryptographic session forks and verifiable assertions.

2. MCP Tool Interfaces

knowledge.retrieve

Retrieves verified facts, session state, and constraints anchored to specific time slices or namespaces.

JSON-RPC Input Schema:

```
{  
  
  "name": "knowledge.retrieve",  
  
  "parameters": {  
  
    "type": "object",  
  
    "required": ["query"],  
  
    "properties": {  
  
      "query": {  
  
        "type": "string",  
  
        "description": "Natural language or keyword target."  
  
      }  
  
    }  
  
  }  
}
```

```
    },  
    "namespace": {  
      "type": "string",  
      "description": "Target memory partition (e.g., 'architecture', 'operator_rules',  
'hardware_limits').",  
      "default": "all"  
    },  
    },  
    "time_slice_proof": {  
      "type": "string",  
      "description": "Optional ISO timestamp or commit hash to retrieve point-in-time memory."  
    },  
    },  
    "max_results": {  
      "type": "integer",  
      "default": 5  
    }  
  }  
}  
}
```

Output Response Format:

```
{  
  "content": [  
    {
```

```

    "type": "text",

    "text": "{\n  \"records\": [\n    {\n      \"entity_id\": \"entity_01\",\n      \"verdict\": \"VERIFIED\",\n      \"assertion\": \"OmniRoute operates as an inference gateway and token compressor on port 20128.\",\n      \"provenance\": \"00_DEFINITIVE_MASTER_SPECIFICATION_V3_COMPLETE.md\"\n    }\n  ]\n}"

  }

]

}

```

knowledge.record

Commits a validated fact, architectural rule, or session fork to the persistent store.

JSON-RPC Input Schema:

```

{

  "name": "knowledge.record",

  "parameters": {

    "type": "object",

    "required": ["assertion", "namespace", "provenance"],

    "properties": {

      "assertion": {

        "type": "string",

        "description": "The exact fact or architectural constraint to persist."

      },

      "namespace": {

```

```
"type": "string",

"description": "Target subsystem partition."

},

"provenance": {

  "type": "string",

  "description": "Source file, commit hash, or operator instruction confirming this assertion."

},

"confidence": {

  "type": "number",

  "default": 1.0

}

}

}

}
```

3. Session Fork & Time-Slice Proof Rules

1. **Deterministic Forks:** Every major agentic session forks with an `ob1_fork_id` referencing the latest ledger commit on Node Beta.
2. **Immutable History:** Assertions once confirmed cannot be deleted; they can only be superseded by a new assertion referencing the older `supersedes_id`.
3. **Conflict Resolution:** If two models assert contradictory facts during a session, OB1 marks the state as `NEEDS_ARBITRATION` and routes it to `UNRESOLVED.md` for human operator review.